

# GraphTheorySymbol Memo

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Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

| Long name                                              | Short name              | Result                                                                                | In context           |
|--------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|----------------------|
| <code>\GTSadjacency/</code>                            | <code>\GTSa/</code>     |    | $u \text{ --- } v$   |
| <code>\GTSadjacencyRelation/</code>                    | <code>\GTSaR/</code>    |    | $u \text{ --- } v$   |
| <code>\GTSunadjacency/</code>                          | <code>\GTSu/</code>     |    | $u \text{ --- } v$   |
| <code>\GTSunadjacencyRelation/</code>                  | <code>\GTSuR/</code>    |    | $u \text{ --- } v$   |
| <code>\GTSdirectedAdjacencyUp/</code>                  | <code>\GTSdAU/</code>   |    | $u \text{ --> } v$   |
| <code>\GTSdirectedAdjacencyUpRelation/</code>          | <code>\GTSdAUR/</code>  |   | $u \text{ --> } v$   |
| <code>\GTSnegatedDirectedAdjacencyUp/</code>           | <code>\GTSnDAU/</code>  |  | $u \not\text{-->} v$ |
| <code>\GTSnegatedDirectedAdjacencyUpRelation/</code>   | <code>\GTSnDAUR/</code> |  | $u \not\text{-->} v$ |
| <code>\GTSdirectedAdjacencyDown/</code>                | <code>\GTSdAD/</code>   |  | $u \text{ --> } v$   |
| <code>\GTSdirectedAdjacencyDownRelation/</code>        | <code>\GTSdADR/</code>  |  | $u \text{ --> } v$   |
| <code>\GTSnegatedDirectedAdjacencyDown/</code>         | <code>\GTSnDAD/</code>  |  | $u \not\text{-->} v$ |
| <code>\GTSnegatedDirectedAdjacencyDownRelation/</code> | <code>\GTSnDADR/</code> |  | $u \not\text{-->} v$ |
| <code>\GTSincidence/</code>                            | <code>\GTSi/</code>     |  | $v \text{ --- } e$   |
| <code>\GTSincidenceRelation/</code>                    | <code>\GTSiR/</code>    |  | $v \text{ --- } e$   |
| <code>\GTSnegatedIncidence/</code>                     | <code>\GTSnI/</code>    |  | $v \not\text{---} e$ |
| <code>\GTSnegatedIncidenceRelation/</code>             | <code>\GTSnIR/</code>   |  | $v \not\text{---} e$ |
| <code>\GTSoutIncidence/</code>                         | <code>\GTSoI/</code>    |  | $v \text{ --- } e$   |
| <code>\GTSoutIncidenceRelation/</code>                 | <code>\GTSoIR/</code>   |  | $v \text{ --- } e$   |
| <code>\GTSnegatedOutIncidence/</code>                  | <code>\GTSnOI/</code>   |  | $v \not\text{---} e$ |
| <code>\GTSnegatedOutIncidenceRelation/</code>          | <code>\GTSnOIR/</code>  |  | $v \not\text{---} e$ |
| <code>\GTSinIncidence/</code>                          | <code>\GTSiI/</code>    |  | $v \text{ --- } e$   |
| <code>\GTSinIncidenceRelation/</code>                  | <code>\GTSiIR/</code>   |  | $v \text{ --- } e$   |
| <code>\GTSnegatedInIncidence/</code>                   | <code>\GTSnII/</code>   |  | $v \not\text{---} e$ |
| <code>\GTSnegatedInIncidenceRelation/</code>           | <code>\GTSnIIR/</code>  |  | $v \not\text{---} e$ |

| Long name                                           | Short name  | Result | In context |
|-----------------------------------------------------|-------------|--------|------------|
| \GTSadjacencyMonochrome/                            | \GTSaM/     |        | u  v       |
| \GTSadjacencyMonochromeRelation/                    | \GTSaMR/    |        | u  v       |
| \GTSunadjacencyMonochrome/                          | \GTSuM/     |        | u  v       |
| \GTSunadjacencyMonochromeRelation/                  | \GTSuMR/    |        | u  v       |
| \GTSdirectedAdjacencyUpMonochrome/                  | \GTSdAUM/   |        | u  v       |
| \GTSdirectedAdjacencyUpMonochromeRelation/          | \GTSdAUMR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpMonochrome/           | \GTSnDAUM/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpMonochromeRelation/   | \GTSnDAUMR/ |        | u  v       |
| \GTSdirectedAdjacencyDownMonochrome/                | \GTSdADM/   |        | u  v       |
| \GTSdirectedAdjacencyDownMonochromeRelation/        | \GTSdADMR/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownMonochrome/         | \GTSnDADM/  |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownMonochromeRelation/ | \GTSnDADMR/ |        | u  v       |
| \GTSincidenceMonochrome/                            | \GTSiM/     |        | v  e       |
| \GTSincidenceMonochromeRelation/                    | \GTSiMR/    |        | v  e       |
| \GTSnegatedIncidenceMonochrome/                     | \GTSnIM/    |        | v  e       |
| \GTSnegatedIncidenceMonochromeRelation/             | \GTSnIMR/   |        | v  e       |
| \GTSoutIncidenceMonochrome/                         | \GTSoIM/    |        | v  e       |
| \GTSoutIncidenceMonochromeRelation/                 | \GTSoIMR/   |        | v  e       |
| \GTSnegatedOutIncidenceMonochrome/                  | \GTSnOIM/   |        | v  e       |
| \GTSnegatedOutIncidenceMonochromeRelation/          | \GTSnOIMR/  |        | v  e       |
| \GTSinIncidenceMonochrome/                          | \GTSiIM/    |        | v  e       |
| \GTSinIncidenceMonochromeRelation/                  | \GTSiIMR/   |        | v  e       |
| \GTSnegatedInIncidenceMonochrome/                   | \GTSnIIM/   |        | v  e       |
| \GTSnegatedInIncidenceMonochromeRelation/           | \GTSnIIMR/  |        | v  e       |
| <hr/>                                               |             |        |            |
| \GTSsetMainColor{blue}                              |             |        |            |
| Long name                                           | Short name  | Result | In context |
| \GTSadjacency/                                      | \GTSa/      |        | u  v       |
| \GTSadjacencyRelation/                              | \GTSaR/     |        | u  v       |
| \GTSunadjacency/                                    | \GTSu/      |        | u  v       |
| \GTSunadjacencyRelation/                            | \GTSuR/     |        | u  v       |
| \GTSdirectedAdjacencyUp/                            | \GTSdAU/    |        | u  v       |
| \GTSdirectedAdjacencyUpRelation/                    | \GTSdAUR/   |        | u  v       |
| \GTSnegatedDirectedAdjacencyUp/                     | \GTSnDAU/   |        | u  v       |
| \GTSnegatedDirectedAdjacencyUpRelation/             | \GTSnDAUR/  |        | u  v       |
| \GTSdirectedAdjacencyDown/                          | \GTSdAD/    |        | u  v       |
| \GTSdirectedAdjacencyDownRelation/                  | \GTSdADR/   |        | u  v       |
| \GTSnegatedDirectedAdjacencyDown/                   | \GTSnDAD/   |        | u  v       |
| \GTSnegatedDirectedAdjacencyDownRelation/           | \GTSnDADR/  |        | u  v       |
| \GTSincidence/                                      | \GTSi/      |        | v  e       |
| \GTSincidenceRelation/                              | \GTSiR/     |        | v  e       |
| \GTSnegatedIncidence/                               | \GTSnI/     |        | v  e       |
| \GTSnegatedIncidenceRelation/                       | \GTSnIR/    |        | v  e       |
| \GTSoutIncidence/                                   | \GTSoI/     |        | v  e       |
| \GTSoutIncidenceRelation/                           | \GTSoIR/    |        | v  e       |
| \GTSnegatedOutIncidence/                            | \GTSnOI/    |        | v  e       |
| \GTSnegatedOutIncidenceRelation/                    | \GTSnOIR/   |        | v  e       |
| \GTSinIncidence/                                    | \GTSiI/     |        | v  e       |
| \GTSinIncidenceRelation/                            | \GTSiIR/    |        | v  e       |
| \GTSnegatedInIncidence/                             | \GTSnII/    |        | v  e       |
| \GTSnegatedInIncidenceRelation/                     | \GTSnIIR/   |        | v  e       |

| Long name                 | Short name           | Result       | In context   |
|---------------------------|----------------------|--------------|--------------|
| <code>\GTSMetaNot/</code> | <code>\GTSMN/</code> | $\mathbb{M}$ | $\mathbb{M}$ |
| <code>\GTSMetaAnd/</code> | <code>\GTSMa/</code> | $\mathbb{A}$ | $\mathbb{A}$ |
| <code>\GTSMetaOr/</code>  | <code>\GTSMO/</code> | $\mathbb{V}$ | $\mathbb{V}$ |

Meta-logic can be used for compound adjacency types, etc.

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u \mathbb{M} \mathbb{V} v$ .

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u \mathbb{M} \mathbb{V} v$ .

Directed graph:

$\forall u, v, u \mathbb{M} \mathbb{V} \mathbb{M} \mathbb{V} v$ .

Oriented graph:

$\forall u, v, u \mathbb{M} \mathbb{V} \mathbb{M} \mathbb{V} v$ .

Tournament adjacency:

$\mathbb{M} \mathbb{V}$  is obtained with `\mathrel{\{\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\}}`.

Tournament adjacency with maybe some colored edge also, who knows?:

$\mathbb{M} \mathbb{V}$  is obtained with `\mathrel{\{\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\}}`.

Directed graph:

$\mathbb{M} \mathbb{V} \mathbb{M} \mathbb{V}$  is obtained with

`\mathrel{\{\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\}}`.

Oriented graph:

$\mathbb{M} \mathbb{V} \mathbb{M} \mathbb{V}$  is obtained with

`\mathrel{\{\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\GTSMO\GTSDAD\GTSDAU\}}`.

Prefix notation:

$\exists u, v, u \mathbb{M} \mathbb{V} v \Leftrightarrow \exists u, v, u \mathbb{M} \mathbb{V} v$ .

$\exists v, e, v \mathbb{M} \mathbb{V} e \Leftrightarrow \exists v, e, v \mathbb{M} \mathbb{V} e$ .

$\mathbb{M} \mathbb{V}$  is obtained with

`\mathrel{\{\GTSMO\GTSxa[mC=red,d]\GTSxa[mC=green,d]\GTSxa[mC=blue,u]\}}`.

$\mathbb{M} \mathbb{V}$  is obtained with

`\mathrel{\{\GTSMO\GTSxi[mC=red,i]\GTSxi[mC=green,i]\GTSxi[mC=blue,o]\}}`.

Apart from these symbol macros, versatile macros to call them all are available with options: `\GTSxa[]`, `\GTSxadj[]`, or `\GTSxadjacency[]` for adjacency symbols, `\GTSxi[]`, `\GTSxinc[]`, or `\GTSxincidence[]` for incidence symbols, `\GTSxml[]`, or `\GTSxmetallogic[]` for meta-logic symbols, `\GTSx[]{}` for all symbols  
 $(\backslash\text{GTSx}[\text{a}] = \backslash\text{GTSxa}[], \backslash\text{GTSx}[\text{i}] = \backslash\text{GTSxi}[], \backslash\text{GTSx}[\text{m}] = \backslash\text{GTSxml}[])$ .  
See Detailed usage section for full syntax of these macros.